UTKARSH GUPTA

Los Angeles, CA | (213) 696-8156 | utkarshgupta2178@gmail.com | linkedin.com/in/utkgupt | uttug.github.io

EDUCATION

University of Southern California Aug 2024 – May 2026
Master of Science in Computer Science *(GPA: 3.9/4.0)*
Relevant Coursework: DL for Robotic Manipulation, Robotic Perception, Autonomous Systems, Robotic Learning

Maharashtra Institute of Technology, Pune Dec 2020 - Aug 2024
Bachelor of Technology in Computer Science and Engineering *(GPA: 3.8/4.0)*

SKILLS

Languages and Platforms: Python, C++, ROS2, CUDA (C++), Linux
Robotic Perception: 3D Data Processing, Sensor Fusion (Camera + LiDAR), 3D Reconstruction, Digital Twins
Planning, Localization and Simulation: Nav2, RRT, PRM, A*, Pose Estimation, EKF, SLAM, Mujoco, NVIDIA Omniverse
Machine Learning: Vision-Language-Action (RT-X, Octo), Segmentation, Classification, Reinforcement Learning
Libraries and Framework: PyTorch, TensorFlow, Keras, OpenCV, SciPy, NumPy, scikit-learn

RELEVANT EXPERIENCE

Research Assistant Oct 2025 - Present
PSI Lab, USC

- Designed **closed-loop simulation evaluation pipelines** in IsaacSim and MuJoCo to assess policy robustness across real2sim scenarios
- Built scalable **scenario deployment frameworks** supporting multiple robot embodiments, enabling parallelized testing with domain randomization and force perturbations
- Integrated heuristic planners and Vision-Language-Action (VLA) models for **scenario-based trajectory evaluation** and large-scale data generation

Research Assistant Nov 2024 - Nov 2025
CPS Vida Lab, USC

- Developed **failure analysis and interpretability techniques** for VLA models to identify task misalignment and unsafe behaviors in autonomous systems
- Implemented **human-in-the-loop evaluation frameworks** to validate low-confidence policy decisions, improving action success rates by **20%** in safety-critical scenarios
- Analyzed task-level performance regressions across scenario distributions to improve robustness and alignment

Robotics Engineering Intern May 2025 - Aug 2025
ERIC Robotics

- Engineered real-time railtrack and 3D safety algorithms, **improving AGV navigation safety by 25%** for dynamic environments
- **Cut AMR initial pose estimation and localization time by 40%** by optimizing pose estimation with LiDAR scans
- Developed and deployed perception and navigation algorithms on **Jetson AGX Orin**, optimizing CUDA-accelerated libraries for real-time performance

PROJECTS

Digital Twin Pipeline via 3D Reconstruction | Digital Twin, 3D Reconstruction, Photogrammetry

- Architected a 3D reconstruction pipeline to generate high-fidelity models of industrial components,
- Generated reconstruction for **partial (180°-250°) image scans** by applying Poisson surface reconstruction

Camera-Only ADAS for Collision Avoidance and Rerouting | OpenCV, PyTorch, YOLOv5, DeepLabV3, StereoSGBM, KITTI Dataset

- Developed a camera-only ADAS with **85% obstacle detection accuracy up to 50 meters** for KITTI stereo dataset
- Fused depth data with YOLOv5 for 3D localization, **enabling real-time rerouting decisions in <100ms**.
- Implemented A* pathfinding on a dynamic grid, **preventing 81% of simulated collisions** and ensuring safety

RL-Based Stealth Navigation in Multi-Agent Environments | Python, Gym, PyTorch, Matplotlib, Stable-Baselines3

- Engineered a multi-agent system and gym environment where agents **achieved an 85% goal success rate** in stealth tasks.